



Programming Fundamentals

Class Assignment # 3

Total Marks: 20

Deadline: 29-5-25: till 11:59 PM

Instructions:

- 1) You have to upload your assignments on CMS.
 - CPP should be combined with files into zip folder.
- 2) All program codes should be written in C/C++. Students should use Visual studio compiler for coding.
- 3) Plagiarism in the work is Professional dishonesty which leads to strict disciplinary action and zero in grading of Assignment.
- 4) **Make your own functions and every function should perform one functionality only.**
- 5) **Use pointer notation rather than array notation.**
- 6) **Don't use of any built-in function related to char array otherwise your task will be cancelled.**
- 7) **Don't use Global variables and string data type.**

Topics: Demonstrate knowledge of Pointers, dynamic memory allocation, function, file handling in C++ in programming (C++).

CLO NO	CLO STATEMENT	PLO	Taxonomy LEVEL
3	Analyze and design solutions for complex real-world problems using appropriate elementary structs of a programming language.	C3	2

Grading Rubric

Parameters	Weights	Remarks
Completeness	50%	Will determine by checking whether your code implements the tasks we ask to implement
Logic	10%	Logic building
Correctness	20%	Incorrect code behavior or not implementing something you are being told to.
Coding Standards	20%	Your code contains comments, well indented and easy to read. Make use of descriptive variable names and user prompts. For function-based programs the code should be organized in as many functions as required and each function contains single task.

Question 1: Highest Voting Person**[10-Points]**

Write a C++ program that allows you to read the local election data from the file to get the candidate's ID and Vote received by the candidate. The program should then store into the file and display each candidate's ID, the number of votes received, and the percentage of the total votes received by the candidate. Your program should also output the total number of votes and the winner of the election.

Note: Follows the function declarations and you can add more helper functions apart from provided the declaration.

Sample Output

Candidate	Votes Received	% of Total Votes
10005	5000	25.91%
20006	4000	20.73 %
30007	6000	31.09 %
40008	2500	12.95 %
80009	1800	9.33 %

Total 19300

The Winner of the Election is 30007

Function Declarations:

- `void Sizeofarrayfromfile(char* filename,int& numCandidates)`
- `void ReadCandidateData(char *filename, int* candidateIds, int* votesReceived, int numCandidates);`
- `int calculateTotalVotes(int* votesReceived, int numCandidates);`
- `void calculateVotePercentages(double* votePercentages, int* votesReceived, int totalVotes, int numCandidates);`
- `int findWinnerIndex(int* votesReceived, int numCandidates);`
- `void displayCandidateInfo(int* candidateIds, int* votesReceived, double* votePercentages, int numCandidates);`
- `void writeToFile(int* candidateIds, int* votesReceived, double* votePercentages, int numCandidates);`
- `int* dynamic(int numCandidates);`

Question 2: Divide and Conquer Approach-using dynamic char array

[10-Points]

The American Equities investment company offers a wide range of investment opportunities ranging from mutual funds to bonds. Investors can check the value of their portfolio from the American Equities' web page. Information about personal portfolios is protected via encryption and can only be accessed using a password. The American Equities company requires that a password consist of characters, 5 of which must be letters and the other 3 digits. The letters and digits can be arranged in any order. American Equities needs a program for their web page that determines whether or not an entered password is valid

Expected Output:**Example 1:**

Enter an 8 character password: rt56AA7q
Password is valid.

Example 2:

Enter an 8 character password: the476NEw
Password is invalid.

Example 3:

Enter an 8 character password: be68moon
Password is invalid.

Example 4:

Enter an 8 character password: \$retrn99
Password is invalid.

Example 5:

Enter an 8 character password: abc12
Password is invalid.

Bonus: Password should be entered in the form of asterisks: +5
